The PING rate is set by the per-synapse weight, not the pool size

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Abstract

The PING rate is a function of the per-synapse inhibitory weight W^{IE} (measured against a fixed release level g_i^*), **not** of the pool size N_I. Sweeping the two independently, the rate-vs-W^{IE} curves for every N_I collapse onto one.

Methods

1. The mechanism, in words. In a PING cycle the E volley recruits I, the I spikes shunt each E cell toward the inhibitory reversal E_i, and E stays quiet until that shunt decays enough for the drive to climb V back to threshold. The cycle's clock, and hence the rate, is that recovery: how far one inhibitory event drives the conductance past the level that silences E, and how long it takes to decay back. This is a *per-event* quantity. Adding more I cells delivers the same shunt through more synapses; it does not change how deep a single event pins a cell or how long the pin lasts. So the rate should track the per-synapse strength, not N_I. The rest of this section makes that precise.

2. The invariant release level. Reduce to one representative E cell, the COBANet membrane of ar003, with the feedforward excitation written as a tonic current I_{ext} as in exp033. This single-cell picture is a heuristic to *motivate* the prediction below, not a proof of it; the full network (§5, Figure 1) is what tests it. When the inhibitory conductance varies slowly the voltage tracks its instantaneous steady state V_∞ (ar003, eqs 11–13; valid in the high-conductance state of a volley, where the open conductances pull the effective membrane time constant below τ_{GABA}). Setting $V_\infty = V_{\text{th}}$ gives the conductance at which the cell is *released* to fire again,

$$g_i^* = \frac{I_{\text{ext}} - I_{\text{th}}}{V_{\text{th}} - E_i}, \quad I_{\text{th}} \equiv g_L(V_{\text{th}} - E_L). \quad (1)$$

with g_L the leak conductance and E_L the leak reversal (ar003′s membrane constants), V_{th} the spike threshold, and E_i the inhibitory reversal. So g_i^* is built from intrinsic constants and the drive alone: **it contains neither N_I nor the per-synapse weight W^{IE}.** It is the level the cycle is pinned to.

3. The cycle clock. Each inhibitory spike adds W^{IE} to g_i, which then decays with τ_{GABA} (ar003′s exponential synapse, eq 8). After a volley peaks at g_i^{pk} (the sum of the fresh kicks), the conductance decays back through g_i^* after

$$T_{\text{sup}} = \tau_{\text{GABA}} \ln \frac{g_i^{\text{pk}}}{g_i^*}, \quad (2)$$

so the gamma period is $T \approx T_{\text{build}} + T_{\text{sup}}$ and $r_E \propto 1/T$. A self-organized loop recruits I only until g_i crosses g_i^* and E shuts off, so the volley overshoots by about one quantum, $g_i^{\text{pk}} \approx g_i^* + W^{IE}$, and

$$T_{\text{sup}} \approx \tau_{\text{GABA}} \ln \left(1 + \frac{W^{IE}}{g_i^*}\right). \quad (3)$$

4. The prediction: collapse onto W^{IE}. Equation (3) has no N_I in it. The suppression, hence the rate, is a function of the per-event jump W^{IE} relative to the invariant g_i^* alone, increasing (rate falling) with W^{IE} and saturating logarithmically. The test is therefore to vary W^{IE} and N_I *independently* and plot the rate against W^{IE}: if (3) holds, the curves for different N_I must lie on top of one another.

5. Simulation. Untrained PING at $N_E = 1024$, $\Delta t = 0.1$ ms, $T = 200$ ms. Per-synapse W^{IE} swept over {0.5, 1, 2, 4, 8, 16} µS (spread held at 10% of the mean) and pool size N_I over {16, 64, 256}, the two varied on an independent grid. Fixed $W^{EI} \sim \mathcal{N}(1.0, 0.1)$ µS and uniform 25 Hz Poisson input; inference only. The biophysical default is $W^{IE} = 2$ µS.

Results

The curves collapse (Figure 1). At each per-synapse weight the three pool sizes agree to within ≈ 0.1 Hz (at the default $W^{IE} = 2$ µS, $r_E = 6.42$, 6.44, 6.45 Hz for $N_I = 16, 64, 256$), and the rate falls smoothly with W^{IE}, from ≈ 9.4 Hz at 0.5 µS to ≈ 4.2 Hz at 16 µS, the decelerating, saturating dependence eq (3) predicts (the precise functional form is not pinned by six points, and turning the cycle period into a per-cell rate would need a participation factor (3) does not supply). The I rates (right panel) collapse the same way. Pool size does not move the rate; the per-synapse weight does.

The lesson for sweeping or training N_I: the rate is governed by the per-synapse W^{IE} measured against g_i^*. Hold W^{IE} fixed and the rate is invariant to pool size; change N_I alone, at fixed per-synapse weight, and nothing moves.

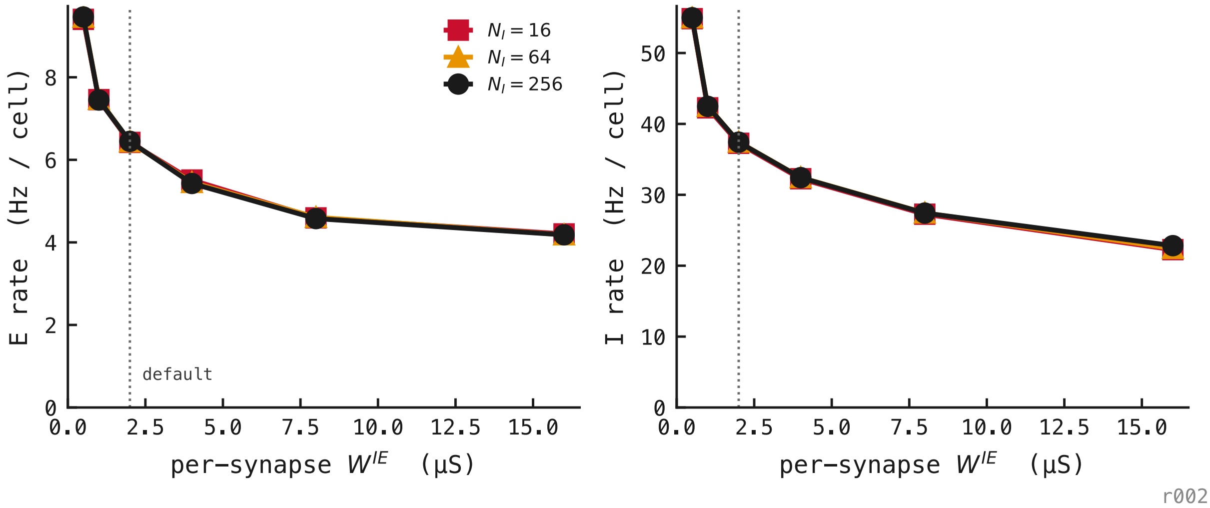

Figure 1: E (left) and I (right) per-cell rate versus the per-synapse weight W^{IE}, one line per pool size N_I. The lines collapse: at any fixed W^{IE} the rate is the same for $N_I = 16$, 64, or 256, while it falls steadily with W^{IE}, the logarithmic, saturating dependence of equation (3). The rate is set by the per-synapse kick, not the pool size. Dotted line marks the biophysical default $W^{IE} = 2$ µS.